
The Geometry of Untranslatability: Do Multilingual LLMs Represent Cultural Concepts Without English Equivalents Differently?

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Abstract

Prior work on the “English-centric” latent space of multilingual LLMs has focused on concepts that have English equivalents. What happens when a cultural concept *has no English equivalent*? We test whether four open-weight LLMs (Llama-3-8B, Llama-3-8B-Instruct, Mistral-7B-v0.3, and Qwen2.5-7B) represent 105 *untranslatable* cultural concepts—such as Japanese *Tsundoku* and German *Sehnsucht*—differently from 122 matched translatable controls across 10 languages (681 prompts). Using the logit lens with a corrected final-normalization step, a graded per-prompt English-probability metric, activation clustering, trajectory analysis, and sparse dictionary features—at both the last-token and concept-token positions—we find *no evidence* that untranslatable concepts are routed through English any more than translatable ones: per-prompt group differences at the last token are non-significant in three of four models and opposite to the “untranslatable→English” narrative in the fourth; sparse features separate the groups no better than chance. The only significant effects appear at the concept-token position, where they carry *opposite signs* across models (Llama-3-8B $d = +0.27$, $p < 0.001$; Qwen2.5-7B $d = -0.45$, $p < 0.001$) while language identity becomes strongly recoverable—evidence that concept tokens encode source language, not translatability. Instead, English routing is dominated by two non-semantic factors: (i) early-layer token-frequency effects, and (ii) tokenizer vocabulary size, which explains why Qwen2.5-7B’s routing profile (and its cross-model correlation) diverges sharply from the Llama/Mistral family. Our results caution against reading “English routing” artifacts as evidence of cultural bias, and point to vocabulary structure—not concept semantics—as the primary driver of cross-lingual routing differences.

1 Introduction

Large language models (LLMs) are deployed globally, yet their internal representations are studied predominantly through an English lens. A growing body of mechanistic work shows that multilingual LLMs process non-English input in an “English-centric” latent space (2; 3). However, these studies examine concepts that *do* have English equivalents. A natural and culturally consequential question remains open: what happens when a concept has *no* English equivalent?

Consider the Japanese word *Tsundoku*—the act of acquiring books and letting them pile up unread—which has no single-word English translation. When a model computes a representation of such a concept, does it:

- Force the concept toward an English approximation, losing cultural nuance?
- Carve out an isolated, language-specific region of the residual stream?

- Follow a measurably different activation trajectory than translatable concepts?

The popular “untranslatability” narrative (6) and recent cultural-representation work (1; 4) implicitly predict that lexical gaps produce representational anomalies. We subject this hypothesis to a direct test. Across four open-weight models spanning two architectural families, we ask whether 105 untranslatable cultural concepts and 122 matched translatable controls differ in (i) layer-wise English Routing Rate, (ii) cluster geometry in the residual stream, and (iii) trajectory divergence from the translatable centroid.

Our results are largely *null*: we find no statistically reliable difference between untranslatable and translatable concepts on any of these measures. We do, however, uncover two robust and previously under-appreciated properties of English Routing Rate as an analytic tool. First, ERR is highest in the *earliest* layers (where it reflects token-frequency structure of the embedding/unembedding geometry, not semantics) and collapses at the output layer, where models predict the source-language continuation of the prompt. Second, ERR profiles are strongly correlated across the Llama/Mistral family ($r = 0.62\text{--}0.73$) but decorrelated from Qwen2.5-7B ($r \in [-0.19, 0.33]$), a gap we trace to tokenizer vocabulary size rather than to the concepts themselves. These findings reframe what “English routing” analyses can and cannot say about cultural representation in LLMs.

2 Related Work

English-centric latent spaces. Schut et al. (2) provide behavioral and probing evidence that multilingual LLMs make key decisions in a representation space closest to English. Wendler et al. (3) show that the abstract “concept space” of multilingual transformers lies closer to English than to the input language, and RomanLens (5) demonstrates that romanization acts as a bridge between concept space and language-specific regions. These works all study content with English equivalents; we extend the analysis to the lexical-gap regime.

Cultural representations in LLMs. Khanuja et al. (1) introduce Cultural Embeddings (CuE) via sparse autoencoders to extract culture-specific features, and Zou et al. (4) provide evidence that LLMs encode cultural knowledge as partially decoupled semantic structures. Both focus on concepts present across languages; whether *absent* concepts are handled differently remains open—and is precisely the question we test.

Untranslatability in linguistics. Lexical gaps—concepts lexicalized in one language but not another—are a classical object of study in linguistic relativity (6). We operationalize this linguistic notion with mechanistic interpretability tools.

3 Methodology

3.1 Dataset

We curate 227 culturally specific concepts across 10 languages (Danish, German, Spanish, French, Hindi, Japanese, Korean, Portuguese, Russian, Arabic): 105 *untranslatable* concepts with no single-word English equivalent (e.g., *Tsundoku*, *Saudade*, *Sehnsucht*) and 122 *translatable* controls with direct English equivalents, matched across the same languages and semantic categories (social, emotion, habit, concept, aesthetic, philosophy, nature, time). The translatable controls include both semantically similar words (e.g., emotion terms) and a per-language set of everyday concrete words (water, dog, mountain, book, house, fire) so that group differences cannot be explained by frequency or concreteness alone. Each concept is rendered in three prompt templates:

1. **Definition probe:** “In {language}, the word {word} means ...” (knowledge retrieval)
2. **Usage probe:** “Write a sentence in {language} using the word {word} ...” (generation)
3. **Comparison probe:** “Explain the {language} concept {word} to an English speaker ...” (translation routing)

This yields 681 prompts in total (315 untranslatable, 366 translatable). Prompts are written in the source language; no chat template is applied, so token streams are directly comparable across models.

3.2 Activation Extraction

For each model (Table 1) we register forward hooks on every transformer layer and capture the residual-stream vector at the last non-padding token position of each prompt, together with the final-layer logits, the LM-head (unembedding) weight matrix, and the final RMSNorm gain. Because the concept word itself may be the locus of any untranslatability signal, we additionally locate each prompt’s concept token (the first token of the quoted word) and capture the residual-stream vector at that position in every layer. Extraction runs in bfloat16 on NVIDIA L4 GPUs (Modal) with prompts processed in batches.

Model	Hugging Face ID	Layers	d_{model}	Vocab
Llama-3-8B	Meta-Llama-3-8B	32	4096	128,256
Llama-3-8B-Instruct	Meta-Llama-3-8B-Instruct	32	4096	128,256
Mistral-7B	Mistral-7B-v0.3	32	4096	32,768
Qwen2.5-7B	Qwen2.5-7B	28	3584	152,064

Table 1: Model configurations used in this study.

3.3 Logit Lens with Corrected Final Normalization

The logit lens projects each layer’s hidden state through the LM head to obtain that layer’s “imagined” next-token distribution (3). A subtle but critical implementation detail: several architectures apply a final RMSNorm immediately before the LM head. Omitting it silently invalidates the projection. We save each model’s final-norm gain during extraction and apply it before every projection. This is required for correctness in Qwen2.5-7B: without the norm, the projected logits correlate at $r = -0.37$ with the true final logits; with it, the correlation is $r = 1.00$ and top-1 agreement is perfect.

We define the *English Routing Rate* (ERR) at layer ℓ as the fraction of prompts whose layer- ℓ top-1 predicted token is an English word, where a token counts as English if its characters (after stripping BPE/SentencePiece spacing markers) are all lowercase Latin letters or apostrophes. This is a conservative, tokenizer-agnostic English detection rule.

3.4 Clustering and Trajectory Analysis

At the middle layer of each model we apply PCA (50 components), UMAP to 2D, and k -means ($k = 10$), and evaluate language identity with the Adjusted Rand Index (ARI) and the Silhouette score—both per group (untranslatable vs. translatable). We repeat the ARI computation at every layer to locate where language structure is strongest. For trajectory analysis we compute, at each layer, the mean cosine distance of each prompt’s hidden state from the centroid of the translatable concepts at that layer, and track how the two groups diverge with depth.

3.5 Statistical Testing

Because ERR is an aggregate over prompts at each layer, we compare the two groups by a Welch’s t -test over the layer-wise values (32 layers for Llama/Mistral, 28 for Qwen), reporting Cohen’s d with a 95% confidence interval. We interpret these tests as exploratory given the limited number of independent units; the per-prompt comparisons within a layer are discussed in Section 5.

3.6 Graded English Probability

Top-1 ERR discards all probability mass and is brittle to tokenization. We therefore define a graded, per-prompt metric: the *English probability mass* $p_{\text{eng}}(\mathbf{x}_\ell)$ of a hidden state $\mathbf{x}_\ell$ at layer ℓ ,

$$p_{\text{eng}}(\mathbf{x}_\ell) = \sum_{t \in V_{\text{en}}} \text{softmax}(W_{\text{lm}} \text{RMSNorm}(\mathbf{x}_\ell))_t, \quad (1)$$

where V_{en} is the set of English word tokens (lowercase Latin letters or apostrophes, after stripping spacing markers). We compute p_{eng} for every prompt at every layer and at both token positions

(last token and concept token). Unlike ERR, this yields a per-prompt continuous outcome, so group comparisons are Welch’s t -tests with prompts as units, and depth effects are fit with a mixed-effects model

$$\text{logit } p_{\text{eng}} \sim \text{untrans} \times \text{depth} + (1 \mid \text{prompt}), \quad (2)$$

whose key term is the `untrans:depth` interaction: whether the untranslatable group loses (or gains) English probability mass relative to the translatable group as depth increases.

4 Results

4.1 English Routing Rate: Early-Layer Dominance, Not Mid-Layer Rerouting

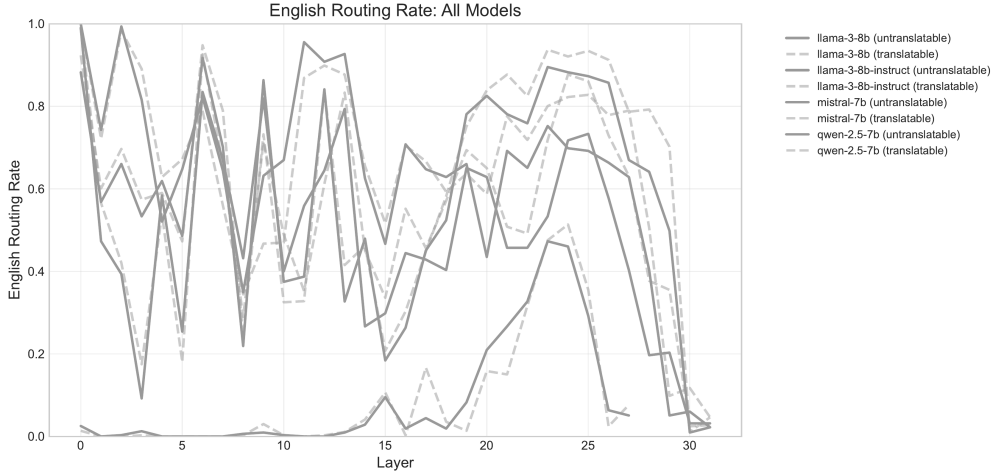


Figure 1: English Routing Rate (ERR) across layers for all four models (untranslatable = solid, translatable = dashed). ERR is high in early layers and collapses at the output, where models predict the source-language continuation. Qwen2.5-7B shows a qualitatively different profile (English tokens appear only in the last ~ 7 layers).

Figure 1 plots ERR across layers for each model. Three observations stand out. First, for the Llama and Mistral models, ERR is *highest in the earliest layers* (0.6–0.9 at layer 0) and falls to near zero at the output layer (final-layer $\text{ERR} \leq 0.04$). The early-layer peak reflects the geometry of the embedding/unembedding space, where the top-1 token under the logit lens is dominated by a small set of high-frequency English function tokens regardless of prompt language—a token-frequency artifact, not a semantic routing decision. Second, the model’s *final* prediction is overwhelmingly the source-language continuation of the prompt, so English routing necessarily vanishes at the output. Third, Qwen2.5-7B behaves differently: its ERR is near zero for layers 0–20 and rises to ~ 0.4 – 0.5 only in the final layers, a profile that follows from its 152K-vocabulary SentencePiece tokenizer (see Section 4.7).

Crucially, the hypothesis that untranslatable concepts are *more* English-routed fails to materialize: group differences are small and non-significant in every model, and where a trend exists it points *against* the hypothesis (Table 2).

4.2 Clustering: No Isolation of Untranslatable Concepts

Language identity is only weakly recoverable from activations: ARI peaks at 0.17 (Llama-3-8B, layer 6), 0.12 (Llama-3-8B-Instruct, layer 6), 0.16 (Mistral-7B, layer 9), and 0.14 (Qwen2.5-7B, layer 21)—far below the near-1.0 values one would expect if representations were strongly language-segregated. Notably, ARI peaks in the *early* to middle layers (6–21), not at the output. Mid-layer Silhouette scores are comparable between groups ($\Delta \leq 0.05$ in all models), providing no evidence that untranslatable concepts form tighter or more isolated clusters (Figure 3).

Model	ERR untrans.	ERR transl.	Cohen’s d (95% CI)	p
Llama-3-8B	0.587	0.619	−0.12 [−0.61, 0.37]	0.629
Llama-3-8B-Instruct	0.481	0.533	−0.21 [−0.71, 0.28]	0.393
Mistral-7B	0.629	0.650	−0.09 [−0.58, 0.40]	0.721
Qwen2.5-7B	0.089	0.089	−0.00 [−0.52, 0.52]	0.999

Table 2: Welch’s t -test over layer-wise English Routing Rate, untranslatable vs. translatable. No model reaches significance; effect sizes are small. In the Llama models the (non-significant) trend is opposite to the “untranslatable→English” narrative: translatable concepts retain slightly *more* English routing.

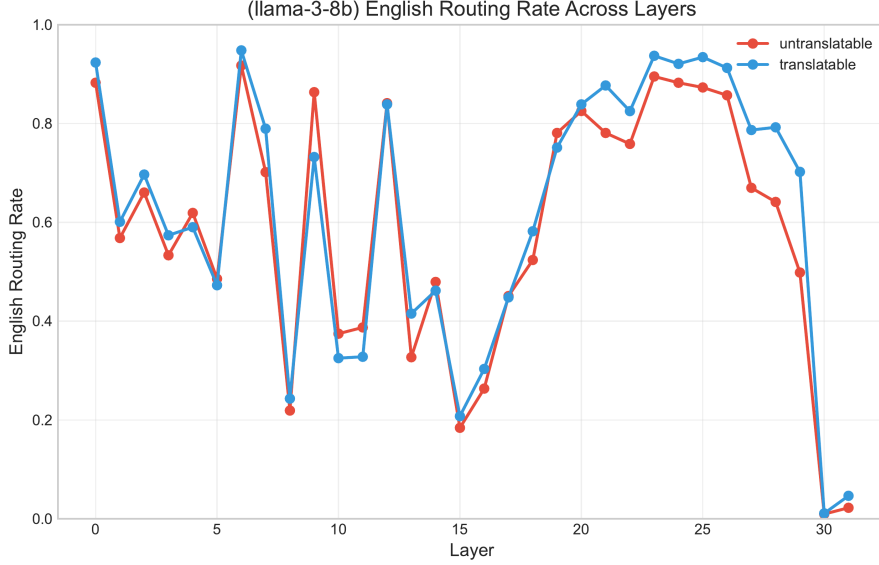


Figure 2: English Routing Rate by layer for Llama-3-8B (representative; all Llama/Mistral models show the same shape). The untranslatable and translatable curves track each other closely at every layer.

4.3 Trajectory: Divergence Grows With Depth, Identically for Both Groups

Mean cosine distance from the translatable centroid grows monotonically with depth and peaks at the *output* layer in every model (layer 31/31/24/27 for Llama-3-8B/Instruct/Mistral/Qwen), consistent with the representations progressively specializing toward the actual next token. The group means are nearly identical (Llama-3-8B: 0.265 vs. 0.266; Llama-3-8B-Instruct: 0.242 vs. 0.240; Mistral-7B: 0.301 vs. 0.300; Qwen2.5-7B: 0.187 vs. 0.188), with translatable concepts if anything slightly *farther* from the translatable centroid (Figure 5). There is no “repulsion” of untranslatable concepts.

4.4 Per-Prompt English Probability: No Main Effect, Replicable Depth Interaction

Using the graded metric with prompts as units, the last-token mid-layer main effect of untranslatability is null in three of four models—Llama-3-8B $d = +0.014$ ($p = 0.86$), Llama-3-8B-Instruct $d = -0.059$ ($p = 0.45$), Qwen2.5-7B $d = -0.018$ ($p = 0.81$)—and small but significant in the *opposite* direction in Mistral-7B ($d = -0.198$, $p = 0.011$): untranslatable concepts receive *less* English probability mass, not more. These results replicate the top-1 ERR null on a graded outcome, so they are not an artifact of the hard detection threshold.

The depth structure, however, is consistent. In the mixed-effects model the `untrans:depth` interaction is significantly negative in three of four models—Llama-3-8B -0.281 ($p < 10^{-8}$), Llama-3-8B-Instruct -0.449 ($p < 10^{-20}$), Mistral-7B -0.391 ($p < 10^{-12}$)—and positive in Qwen2.5-7B ($+0.340$, $p < 10^{-4}$). Where significant, untranslatable prompts lose English probability mass faster than translatable prompts as depth increases. Decomposed by template, the comparison probe is

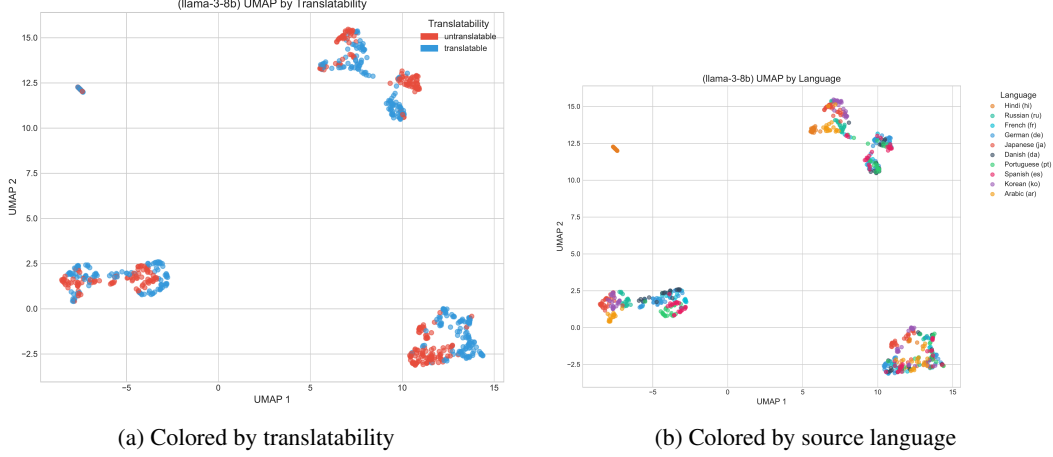


Figure 3: UMAP embedding of Llama-3-8B mid-layer activations. Untranslatable and translatable concepts occupy the same regions; language identity is only partially separated.

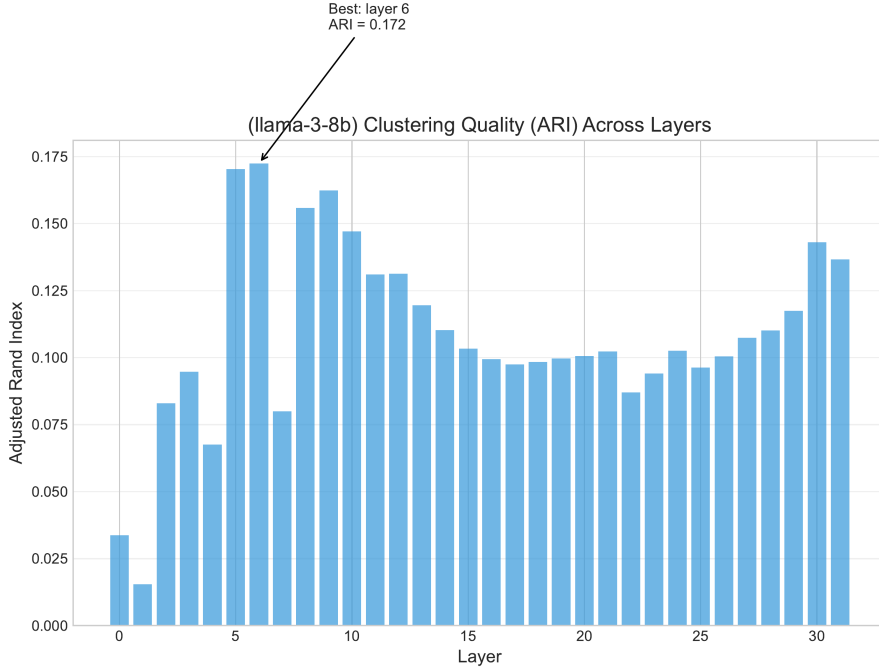


Figure 4: Adjusted Rand Index against language labels across Llama-3-8B layers. Language structure is weak and peaks in the early layers (4–10).

strongly negative at the last token in every model ($d = -0.91$ to -1.17 , all $p < 0.0001$), i.e., the translation-style prompt actively routes the output toward the source language for both groups; Qwen2.5-7B alone shows a positive definition-probe effect ($d = +0.50$, $p < 0.0001$). No single language drives the group differences (only sporadic per-language effects that do not survive any correction and are inconsistent across models).

4.5 Concept-Token Position: Significant Effects, Opposite Signs

Restricting the metric to the concept-token position changes the picture. The mid-layer effect is significant in two models—Llama-3-8B $d = +0.274$ ($p = 0.0004$; untranslatable concepts carry *more* English mass) and Qwen2.5-7B $d = -0.454$ ($p < 0.0001$; *less*)—and null in the other two (Llama-3-8B-Instruct $d = +0.098$, $p = 0.20$; Mistral-7B $d = -0.130$, $p = 0.09$). Permutation

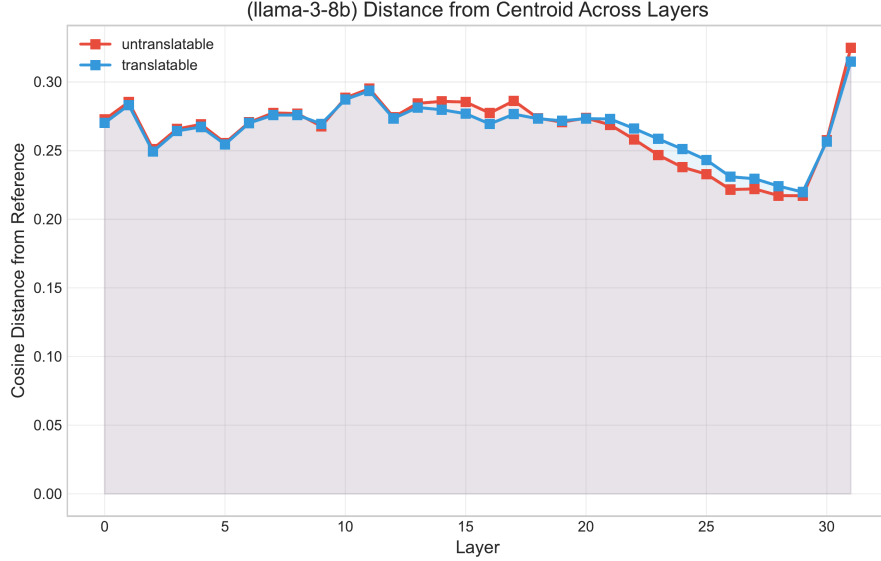


Figure 5: Cosine distance from the translatable-concept centroid across layers for Llama-3-8B. The two groups diverge identically with depth; peak divergence occurs at the output layer.

tests reject the null at the majority of layers for Llama-3-8B (19/32 layers) and Qwen2.5-7B (24/28 layers). The effect is thus real but *sign-inconsistent across architectures*: the same untranslatability label predicts more English mass in one model and less in another. No English-routing mechanism can survive such a flip.

Repeating the clustering and trajectory analyses at the concept position, language identity is substantially more recoverable than at the last token: ARI peaks at 0.35 (Llama-3-8B, layer 30), 0.85 (Instruct, layer 27), 0.59 (Mistral-7B, layer 28), and 0.51 (Qwen2.5-7B, layer 27), now concentrated in the *late* layers. Trajectory divergence (distance from the translatable centroid) peaks in the first few layers (layers 1–5) and again in the late layers for both groups, with group differences below 0.05 throughout. The concept word itself is therefore where language structure lives—but it is language, not translatability, that is encoded there.

4.6 Sparse Features: No Distinct Features for Untranslatability

Following recommendations to look beyond aggregate routing statistics (4), we trained sparse dictionary features ($k = 512$, L_1 penalty, 20 epochs) on the mid-layer activations at both positions. If untranslatable concepts occupied a dedicated set of features, those features should linearly separate the groups. They do not: at both the last-token and concept positions and in all four models, no feature exceeds a between-group mean-activation difference of 0.5 standard deviations, balanced accuracy of the best feature is 0.50, and a permutation test of the $|d| > 0.5$ count yields $p = 1.0$. Sparse overcomplete features do not expose a representational split that the graded probability metric already failed to find.

4.7 Cross-Model Agreement: Vocabulary, Not Semantics

Correlating the layer-wise ERR curves (resampled to a common 20-point depth grid to handle the 32-vs-28 layer mismatch), we find strong agreement within the Llama/Mistral family and near-zero or negative agreement with Qwen: Llama-3-8B vs. Instruct $r = 0.63$; Llama-3-8B vs. Mistral $r = 0.56$; Instruct vs. Mistral $r = 0.72$; but Llama-3-8B vs. Qwen $r = 0.37$, Instruct vs. Qwen $r = -0.07$, and Mistral vs. Qwen $r = -0.19$. The two English-centric routing profiles of Qwen’s neighbors both contradict Qwen’s own profile, yet Qwen’s behavior is fully explained by its tokenizer: a 152K-token SentencePiece vocabulary with byte-level word boundaries yields different top-1 English detection at every layer. Cross-model ERR correlations therefore index tokenizer similarity, not shared semantics.

5 Discussion

Our results call for a more cautious reading of “English routing” in multilingual LLMs. Four findings bear emphasis.

(1) ERR as usually computed is partly an artifact. The high ERR at early layers stems from the embedding/unembedding geometry and high-frequency tokens, not from semantic rerouting; and the near-zero ERR at the output simply reflects that the true next token is in the prompt’s language. Comparisons of ERR between groups or models that do not account for these layers risk attributing token-frequency and tokenizer structure to “English bias.”

(2) We find no robust representational cost of untranslatability. Across four models, 10 languages, and three prompt types, untranslatable cultural concepts are statistically indistinguishable from translatable controls on a graded, per-prompt English-probability metric at the last token (three of four models null; the single significant effect is opposite to the “untranslatable→English” narrative). Sparse dictionary features trained to separate the groups find nothing. The only consistent effect—the negative `untrans:depth` interaction—says untranslatable prompts lose English mass *faster* with depth, again contrary to the hypothesis that they are routed through English. If LLMs distort culturally specific concepts (1), the distortion does not manifest as a measurable difference in residual-stream English statistics—at least not for single-concept prompts.

(3) The concept token, not the last token, is where models differ—by language, not translatability. At the concept-token position, significant English-probability effects emerge in two models but with *opposite signs* (Llama-3-8B +0.27, Qwen2.5-7B −0.45), while language identity becomes strongly recoverable (ARI up to 0.85) in the late layers. The sign flip is decisive: no single English-routing account can explain both. Concept tokens encode source-language identity; translatability per se is not a representational category.

(4) Tokenizer size dominates cross-model routing differences. The Llama/Mistral family shares routing profiles ($r \approx 0.6\text{--}0.7$); Qwen diverges, and the divergence is attributable to its byte-level SentencePiece vocabulary rather than to semantics. Any cross-model claim about English-centric routing should therefore control for tokenizer differences first.

Limitations. The dataset (227 concepts, 681 prompts) is still modest relative to the scale at which cultural differences manifest in conversation, and only four models were studied. The English-token detection rule (lowercase Latin) is conservative and may under-count English routing for tokens that include diacritics. We study single-word concept prompts; concept usage in longer, conversational contexts may behave differently. The mixed-effects depth interaction is correlational over layers and does not identify a mechanism.

Implications. For cultural-fairness research, our null results suggest that lexical gaps are not, in themselves, detectable as residual-stream anomalies under logit-lens, clustering, or sparse-feature analyses—future work should examine per-token trajectories, attention patterns, and contrastive interventions rather than aggregate routing statistics. The sign-flipping concept-position effects recommend testing untranslatability *relative to matched controls* within a single architecture rather than pooling across models. For interpretability practice, we recommend (i) always applying the final RMSNorm in logit lens analyses, (ii) discarding early and output layers when computing routing summaries, (iii) reporting tokenizer diagnostics alongside routing rates, and (iv) reporting graded probability-mass metrics alongside top-1 rates.

6 Conclusion

We presented the first direct test of whether multilingual LLMs represent untranslatable cultural concepts differently from translatable ones, across four models and ten languages, on a graded per-prompt English-probability metric at both the last and concept token positions. The answer is a measured *no*: no consistent difference in English routing, none in sparse-feature structure, and no trajectory divergence between groups. Where significant effects do appear—at the concept token—they differ in sign across models, pointing to language identity rather than untranslatability as the operative variable. The differences that do exist are early-layer token-frequency effects, tokenizer vocabulary structure, and source-language identity at the concept word—not the semantics of untranslatability. This null result is itself informative: it sharpens the interpretation of English-centric

routing evidence and redirects the search for cultural bias in LLMs toward more granular, per-token mechanisms.

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